Truncation

* methods to remove extra bits to create a 24 bit / 53 bits no. that approximates the longer version.
* guard bits — extra bits which are considered in intermediate steps of arithmetic operations to increase the accuracy in the final results.
* removing guard bits — truncation

I — chopping

II — Von Neumann rounding

III — Rounding.

Eg: Consider mantissa size = 3bits

chopping — Remove the guard bits

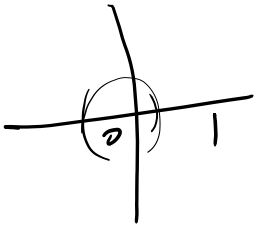
$0.b_{-1}b_{-2}b_{-3}\underbrace{000}_{\text{guard bits}} \Longrightarrow 0.b_{-1}b_{-2}b_{-3}$

$0.b_{-1}b_{-2}b_{-3}$ 111 $\Rightarrow 0.b_{-1}b_{-2}b_{-3}$

guard bits

→ the error ranges from 0 to 1

* error in 3 bit result ranges from 0 to 1 at the lsbit of retained bits
* error change is not symmetric about 0, hence it is a biased approximation.



ii) Von neumann rounding.

* if guard bits are all 0's, then drop the guard bits without changing the retained bits.

Eg: $0.b_{-1}b_{-2}b_{-3}000 \Rightarrow 0.b_{-1}b_{-2}b_{-3}$

$0.1011000 \Rightarrow 0.1011$

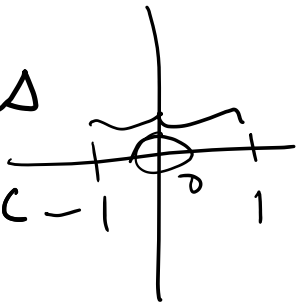
* else if any of guard bits is 1, then set the lsbit of retained bits to 1

Eg: $0.b_{-1}b_{-2}b_{-3}010 \Rightarrow 0.b_{-1}b_{-2}1$

Eg:- $1.10\underbrace{0}_{\text{guard bits}}\underbrace{100}_{} \Rightarrow 1.101$

$1.10\underbrace{1}_{\text{guard bits}}\underbrace{001}_{} \Rightarrow 1.101$

* error change ranges from -1 to 1 at the lsb position of retained bits.
* unbiased approximation as error range is symmetric about 0.



iii) Rounding

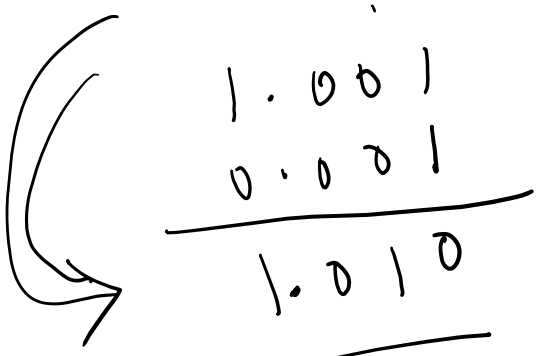
* achieves closest approximation to the original number.
* if msb of guard bits is 1 then add 1 to the lsb of retained bits.

Eg: $1.b_{-1}b_{-2}\underbrace{b_{-3}}_{\text{lsb}}\underbrace{1}_{\text{msb}}01$

0.000 (guard bits)

$\Rightarrow 1.b_{-1}b_{-2}b_{-3} + 0.001$

Eg:- 1.001101



$$\begin{array}{r} 1.001 \\ 0.001 \\ \hline 1.010 \end{array}$$

* error range is $-\frac{1}{2}$ to $+\frac{1}{2}$.
* Symmetric about 0, so unbiased approximation.
* best method
* complex as addition is required
* possible renormalization.

Eg: $1.\underbrace{111}_{\text{1 1 1}}100$

$$\begin{array}{r} 1.111 \\ +\ 0.001 \\ \hline 10.000 \end{array} \Rightarrow 1.0000 \times 2^1$$

1. Convert the decimal fraction 0.1 to a binary fraction. If the conversion is not exact, give the binary fraction approximation to 8 bits after binary point using each of three truncation methods.

$(0.1)_{10} \Rightarrow (\quad)_2$

$0.1 \times 2 = 0.2$
$0.2 \times 2 = 0.4$
$0.4 \times 2 = 0.8$
$0.8 \times 2 = 1.6$
$0.6 \times 2 = 1.2$
$0.2 \times 2 = 0.4$
$0.4 \times 2 = 0.8$
$0.8 \times 2 = 1.6$
$0.6 \times 2 = 1.2$
$0.2 \times 2 = 0.4$

$= (0.00011001 | 10011001\ldots)$

retained | guard bits

1) Chopping:

0.00011001

2) Von Neumann rounding

0.00011001

3) Rounding

$$\begin{array}{r} 0.00011001 \\ +\ 0.00000001 \\ \hline 0.00011010 \end{array}$$

Arithmetic operations

1. add / sub ⇒ opr¹, S_A, S_B

$|2-1| = 1 = n$

Eg: $65.75 \times 10^2 + 12.34 \times 10^1$

$65.75 \times 10^2 + 1.234 \times 10^2$

$+65.750 \times 10^2$
$+\ 1.234 \times 10^2$
$+66.984 \times 10^2$

A+B

$-(M_A)$
$+\ -(M_B)$
$=\ (\ )$

A+B

$-\ M_A$
$+\ M_B$

A⊕B

$-\ M_A$
$+\ -M_B$
$=\ (\ )$

1) Add -1.001×2^{-2} & -1.111×2^{0}

A B

$E_A = -2 \quad E_B = 0$

$|-2-0| = 2$ bits

* M_A should be shifted right by 2bits

$A = -0.01001 \times 2^{0}$

$+ B = -1.11100 \times 2^{0}$

-10.00101×2^{0}

$1.0\,00 \mid 1\,01 \times 2^{1}$; assume 3 bits mantissa size

$+ \quad 0.0\,01$ guard ; use rounding

-1.001×2^{1}

Result $\Rightarrow$ -1.001×2^{1}

$X = -1$

$X = 0$

~~$X = -2$~~

Eg1) Add A & B in SP format

A = 44900000H, B = 42A00000H

	S	E'	M
A =	0	1000 1001	001 0000 0000 ...
B =	0	1000 0101	010 0000 0000
		0000 0101	0000 0000

↳ E+127

$E'_A = 137 = 128 + 8 + 1 = 10001001$

$E'_B = 133 = 128 + 4 + 1 = 10000101$

$n = |137 - 133| = 4$

Shift M_B right by 4 bits

$M_A = 1.0010000\ldots0$

$M_B = 1.0000010\ldots0$

$+10.0010010\ldots0 \times 2^{137}$

$1.0001001\ldots0 \times 2^{138}$

Res = | 0 | 10001010 | 00010010...0 |

(0 → +ve; 10001010 → 138; 00010010...0 → M)